

Trainer Debrief — Lab 23

Full Factorial Design of Experiments (Black Belt Only)

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Design, run and analyse a full factorial experiment on your capstone process (A3, A5, K2).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-23-full-factorial-doe.xlsx

Main effects: ClampPressure +18.8 kPa ($p < 0.001$), DwellTime +14.2 kPa ($p < 0.001$), AmbientTemp -4.0 kPa (not significant, $p \sim 0.09$). The AB (clamp x dwell) INTERACTION is +11.6 kPa and highly significant ($p < 0.001$) — this is why one-factor-at-a-time would have missed the optimum. R-sq ~ 0.97 . Best setting: clamp HIGH (46 bar) and dwell HIGH (2.7 s), predicted Y ~ 240 kPa, comfortably centred in the 180-260 spec. Temperature can be left uncontrolled. Carry these settings into Labs 24-25.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Your design is fully randomised with replication, you have computed and plotted both main effects and interactions, effect significance is assessed statistically, and confirmation runs validate the predicted optimum.

Where this sits in the capstone

BLACK BELT ONLY — the Green Belt course mentions DOE in passing; the Black Belt owns it. One-factor-at-a-time experimentation is slower, less informative and structurally incapable of detecting interactions. Full factorial designs estimate every main effect AND every interaction. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27

Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30
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If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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